



CALICUT UNIVERSITY – FOUR YEAR UNDER GRADUATE PROGRAMME (CU-FYUGP)

B.Sc. PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	OPTICS AND SPECTROSCOPY				
Type of Course	Minor (GROUP V: ENERGY PHYSICS)				
Semester	III				
Academic Level	200 - 299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Basics of Physics and Chemistry (Plus Two Level)				
Course Summary	This course explores the fundamental properties of light, its interaction with matter, and spectroscopic techniques used to analyze molecules.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Explain the laws of reflection and refraction, and how they influence light behavior.	U	F	Instructor-created exams / Quiz
CO2	Describe the electromagnetic spectrum and differentiate between wave and particle properties of light.	U	C	Seminar Presentation / Group Tutorial Work
CO3	Analyze the principles of interference and apply them to phenomena like Young's double slit experiment.	An	P	Practical Assignment / Observation of Practical Skills
CO4	Explain the concept of polarization and apply it to phenomena like Brewster's Law.	Ap	P	Instructor-created exams /

				Home Assignments
CO5	Discuss the principles of optical activity and how it relates to specific rotation.	Ap	C	Practical Assignment / Observation of Practical Skills
CO6	Explain the fundamental concepts of spectroscopy, including energy quantization, absorption/emission, and different spectroscopic methods like microwave and infrared spectroscopy.	An	P	Instructor created exam/Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	Introduction		8	15
	1	Properties of light, Laws of reflection, laws of refraction	3	
	2	Refractive index, Optical path	2	
		Electromagnetic spectrum and visible light	1	
	3	Photons, Dual nature of light	2	
	Sections 1.5 – 1.12, Book 1			
II	Interference and Polarization		19	25
	4	Interference, Young double slit experiment	2	
	5	Coherence and conditions for interference	2	
	6	Interference in thin parallel films	3	
	7	Interference in wedge shaped film, Angle of wedge and thickness of spacer,	3	
	8	Colour of thin films		
	9	Polarization: Types of polarization	1	
	10	Brewster's law	1	
	11	Production of plane polarized light	2	

	12	Polarizer and analyser, Malu's law	2	
	13	Double refraction	1	
	14	Optical activity and specific rotation	2	
	Section 14.4 – 14.7, 15.2, 15.5, 20.1, 20.2, 20.5, 20.6, 20.8 - 20.11, 20.27 - 20.29 , Book 1			
III	Introduction to Spectroscopy		7	15
	15	Electromagnetic spectrum and Quantization of energy	2	
	16	Types of molecular energies and spectroscopic methods	2	
	17	Spectral line width	1	
	18	Absorption and emission of radiation, Einstein coefficient (excluding derivation)	2	
	Sections : 1.1 - 1.6, Book 2			
IV	Spectroscopic Methods of sample analysis		11	15
	19	Microwave spectroscopy	3	
	20	Infrared Spectroscopy (vibration spectra only)	2	
	21	Electronic spectroscopy	3	
	22	Raman spectroscopy: Introduction, Quantum theory of Raman scattering, Rotational Raman spectra of linear molecules	3	
	Sections 8.6 - 8.8, Book 3, Sections 8.1, 8.2.2 , 8.3.1, Book 2			
V	PRACTICALS		30	
	1	Determine the focal length of the combination of two lenses separated by a distance. - Determine the focal lengths, f_1 and f_2 of the two lenses using an illuminated cross-slit screen holder, nodal slide(for placing the lenses) and plane mirror arrangement. - Place the two lenses separated by a distance d, determine the focal length, F of the combination and verify the relation $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$. - The combination of the lenses in the eyepiece of the spectrometer/ travelling microscope may be used for the study. https://www.youtube.com/watch?v=IOIEEtyNPBg		

		• https://www.youtube.com/watch?v=tNo4Ipk74SU		
	2	Determine the refractive index of (a) given liquid and (b) the material of a lens, by forming a liquid lens. • Through this experiment the students are expected to get the concepts of image formation, combination of lenses and radius of curvature of the surface of lens. • Determine the radius of curvature of the lens by Boy's method and hence calculate the refractive indices.		
	3	Determination of the dispersive power of a solid prism using a spectrometer. • Find the angle of the prism and the angle of minimum deviation for prominent lines of the mercury spectrum using a spectrometer. • Calculate the refractive indices corresponding to the colors and find the dispersive power of the material of the prism for two pairs of wavelengths.		
	4	Refractive indices of quartz prism using spectrometer. • Determine the refractive indices of quartz for the ordinary and extraordinary rays of a sodium vapour lamp by arranging the quartz prism at minimum deviation position in the spectrometer. • Verify the polarizations of the ordinary and extraordinary rays using a polaroid.		
	5	Determination of wavelengths of mercury spectrum using diffraction grating and spectrometer. • Arrange the grating at normal incidence. • Standardize the grating using the green line of mercury and then find the wavelengths of other prominent lines of the spectrum.		
	6	Newton's rings-determination of the wavelength of sodium light • Form of Newton's rings in the air-film in between a plano-convex lens and a glass plate using sodium-source. • Determine the radius of curvature by Boy's method and determine the wavelength of the source. • Optional: In experiment 5 and 6, record a short video of the interference pattern, calibrate the video using scale marked on the glass plate, analyse the video using Tracker tool. From the		

		intensity profile get the locations of the dark rings and calculate the wavelength of the source/thickness of the sample https://physlets.org/tracker/. https://www.youtube.com/watch?v=UCCPkJpUQEw		
	7	Air wedge-determination of the radius of a thin wire/human hair/thin foil. - Form interference fringes using sodium-source, in the air-film in between wedge formed by placing the given sample between the glass plates. - Measure the positions of the successive dark bands using a travelling microscope and determine the angle of the wedge and thickness of the sample given.		
	8	Single slit diffraction using laser - Determination of slit width. - The laser light diffracted from the narrow slit is allowed to fall on a screen and record the maxima or minima points in a paper. - From the width of the central maxima or the position of minimum intensity points, calculate the slit width. - Verify the slit width using a traveling microscope. - Wavelength of laser can be found using diffraction grating of known N.		
	9	Study the specific rotation of the sugar solution using a polarimeter. - Determine the specific rotation corresponding to different concentrations of the sugar dissolved in water. - Draw a graph between rotation and concentrations and verify the linear relationship.		
	10	Verification of Malus's law using polarizer, analyzer and photo detector - Unpolarized light is allowed to pass through a polarizer and is observed through an analyzer. - Vary the angle between the axes of polarizer and analyzer and measure the intensity of the light (current output of the photodetector). - Plot $\theta - I$ and $\cos^2 \theta - I$ graphs and verify the Malus's law.		

		• A flat computer monitor (or LCD TV screen) in plain white color can be used as the source of linear polarized light. • The ambient light sensor of the smartphone and the orientation sensor of the smartphone can be used to measure the illuminance and the angles respectively. • A small piece of polarizer (a square of about 1 cm side) from an old calculator's display was placed over the ambient light sensor as analyser. • https://arxiv.org/pdf/1607.02659		
	11	Spectrometer-Determination of the Cauchy's constants of the given prism • Find the angle of the prism, the minimum deviation angles of the prominent lines of the mercury spectrum and hence calculate the refractive indices for the colors. • Determine A and B from the $\mu - \frac{1}{\lambda^2}$ graph.		
	12	Determine the numerical aperture (NA) of an optical fiber using a laser • Couple the light from the laser source onto one of the fiber ends and the light coming from the other end is allowed to fall on a screen(sheet having circular markings) placed perpendicular to the axis of the fiber. • Measure the diameter of the laser beam on the screen and the distance between the screen and fiber output end and hence calculate the NA.		
	13	Determination of Plank's constant using LEDs • Observe the turn-on voltage, • V_0 of LEDs and calculate the value of h. Use at least 4 different colors of LED (with transparent casing) • Plot $\frac{1}{\lambda} - V_0$ graph using Python, fit a straight line to get the slope and estimate the value of h. • Calculate the %error. • Programmable voltage source of ExpEYES may be used to find the turn-on voltage.		

14	Continuous and line spectra- Determination of the wavelengths and photon energy. - Familiarize the initial adjustments and measurements in the spectrometer. - Mount the grating at normal incidence on the spectrometer. - Determine the wavelengths of the sodium vapor lamp and calculate the associated photon energy. - Determine the approximate range of the wavelengths of the continuous spectrum of incandescent/white LED lamp or any one coloured LED and calculate the associated photon energy. - The readings of the first order spectrum will be enough. Number of lines/m of the grating can be given.		
15	Analysis of Hydrogen spectra using the Tracker Video Analysis tool. - Calibrate the video of the Hydrogen spectra in the Tracker tool using two laser wavelengths/lines of mercury spectra. - Plot the intensity profile, find the prominent wavelengths of the Balmer series and calculate the Rydberg's constant. - Estimate the %error. - Pre recorded video of the Hydrogen spectra can be used. https://physlets.org/tracker/. https://www.youtube.com/watch?v=UCCPkJpUQEw		

Books and References:

1. A Textbook of Optics by N. Subramanyam, Brij Lal, M N Avadhanulu 25TH Edition (Book 1)
2. Molecular structure and spectroscopy, (Second edition) by G. Aruldas (Book 2)
3. Concepts of Modern Physics by Arthur Beiser, 6th edition
4. Optics by Eugene Hecht
5. Fundamentals of - Molecular Spectroscopy - THIRD EDITION, by C N Banwell

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	2	0	0	0	2	0	2	0	0	1	1	0	0
CO 2	2	0	0	0	2	0	2	0	0	1	1	0	0
CO 3	2	1	1	0	2	0	2	0	0	1	1	0	0

CO 4	2	0	1	0	2	0	2	0	0	1	1	0	0
CO 5	2	0	1	0	2	0	2	0	0	1	1	0	0
CO 6	2	1	2	0	2	1	2	0	0	1	1	0	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/ Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	

VOCATIONAL MINOR COURSES